DATA RIDE



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Data Ride uses **open-source technology** and **cyclist-generated data** to reveal real cycling conditions and infrastructure gaps in the Lorraine region. The project starts by engaging local stakeholders in Nancy through a co-created mapping, while a senseBox:bike prototype is developed in Münster.

We then run a **data collection campaign** and transform the results into visual material to support a **local advocacy strategy**. Finally, we launch the advocacy campaign together with partners and produce collaboratively a policy recommendations document.

The project concludes with advocacy sustainability guidelines and the **Data Ride Advocacy Toolkit**, designed for broader dissemination and long-term inspiration.

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overview

Nancy’s landscape of cycling practices mapping

Contact main actors → 'Snowball'to enrich the mapping

Validate hypotheses that can be dada-tested

Organise who would be up to collect this data, cycling with senseBOX →

In parallel: Software/Hardware prototype development in Münster and tested in Germany

Publish of the Tech. materials in Open Source license

Installation of the prototype in the bikes and data collection - Community Sensing Campaign with local cyclists

Data analysis and collective preparation of the advocacy campaign with local actors

We are here!

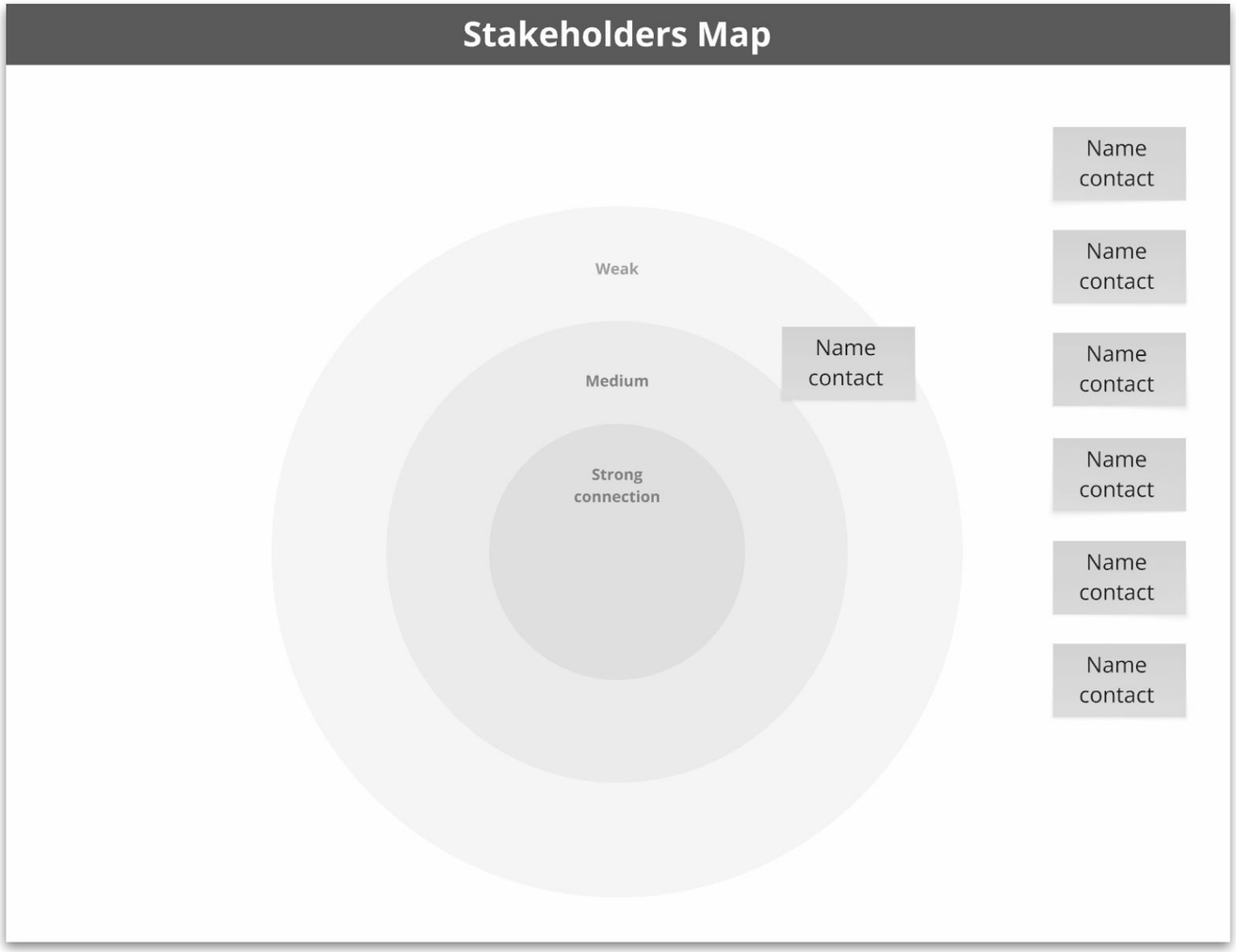
Development of the advocacy campaign strategy, including communication materials such as posters, stickers, online reach, politicians contacts

Campaign kick-off and preparation of a policy recommendations' draft →

Advocacy sustainability plan with motivated local actors, final policy recommendations document →

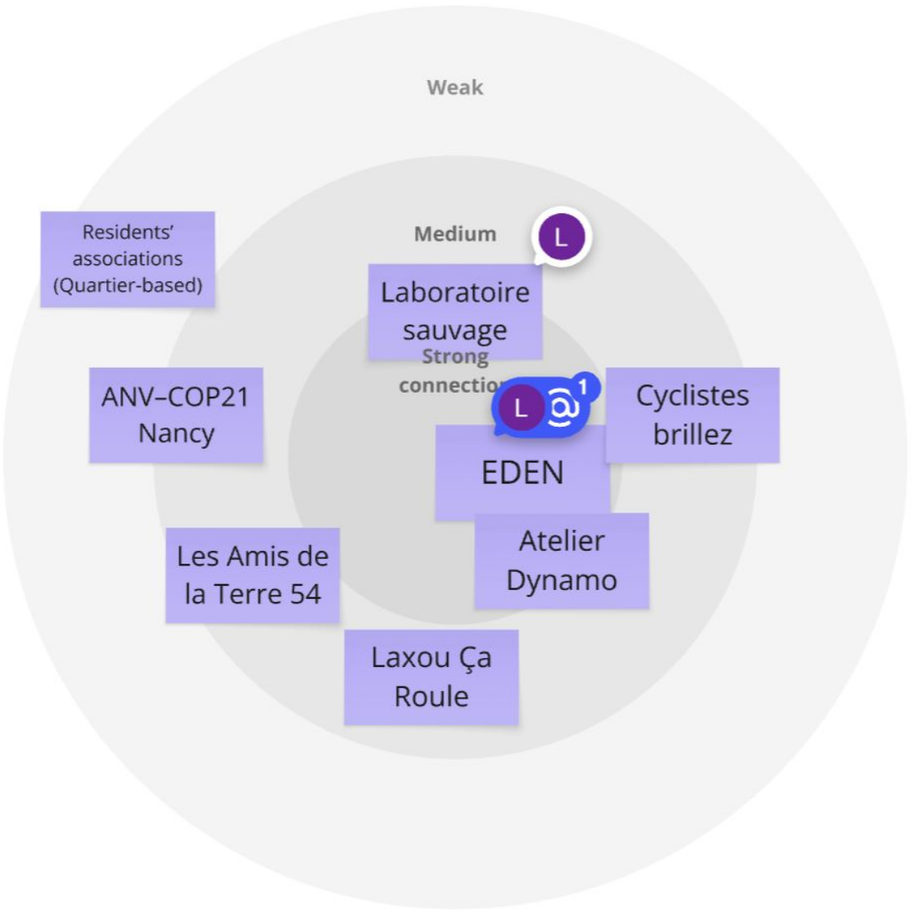
Data Ride Advocacy toolkit for dissemination and inspiration

Nancy’s landscape of cycling practices



https://miro.com/app/board/uXjVJuWNhVw=/

Citizen-led groups (Neighbourhood groups, associations, NGOs, etc)



Name contact

Name contact

Name contact

Name contact

Name contact

Installation of the prototype in the bikes and data collection

+

Community Sensing Campaign with local cyclists

Device installation





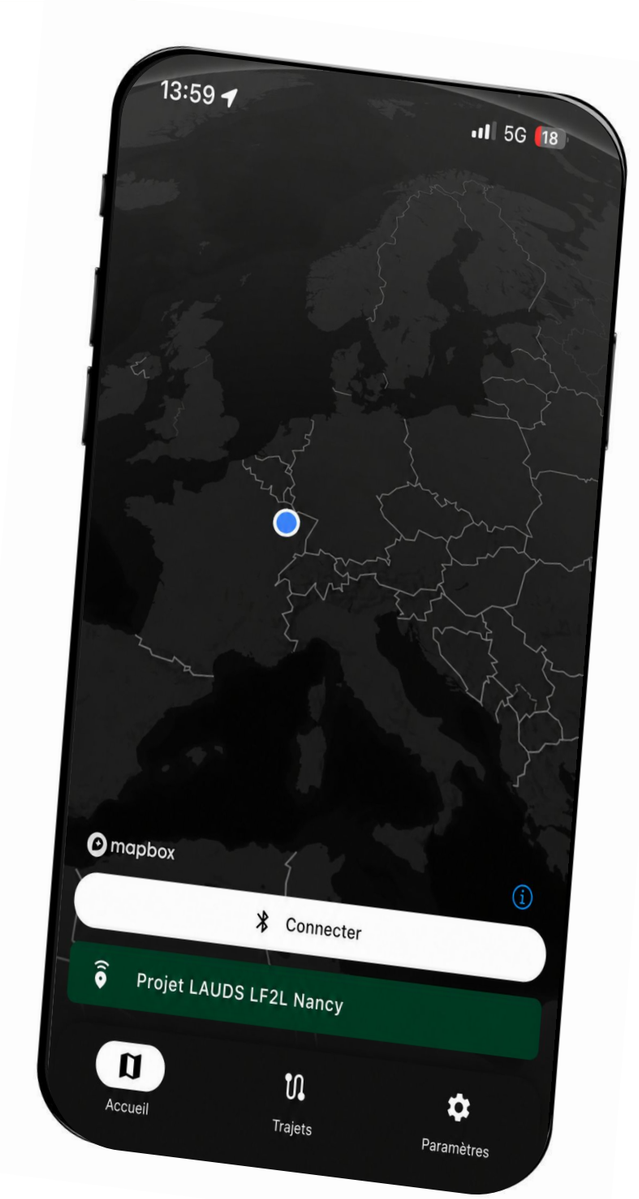


Mounting system improvements - 3D printing





Software update in French



Initial discussion with mapped stakeholders + Focus Group



Updated Hypothesis

H1 · The continuity and design quality of cycling infrastructure are decisive to adopting bikes as a means of transport

The adoption and everyday use of cycling depend less on the existence of isolated bicycle lanes and more on the continuity, width, and design quality of a coherent cycling network.

Discussions with cyclists and local stakeholders highlighted several issues affecting perceived safety and usability:

- discontinuous cycling lanes
- inconsistent lane markings
- narrow infrastructures squeezed between parked cars and traffic
- bicycle lanes placed on streets where spatial constraints create uncomfortable or unsafe riding conditions

Updated Hypothesis

H2 · Cycling safety emerges from interactions between mobility modes

Cycling safety is shaped not only by infrastructure but also by dynamic interactions between cyclists, cars, pedestrians, and micro-mobility users.

Focus group discussions highlighted multiple sources of perceived danger:

- proximity between cars and bicycles
- pedestrian behaviour in shared spaces (particularly near tram lanes)
- unpredictability of scooters and other micromobility devices
- signalling and awareness gaps among drivers (especially in crossroads)

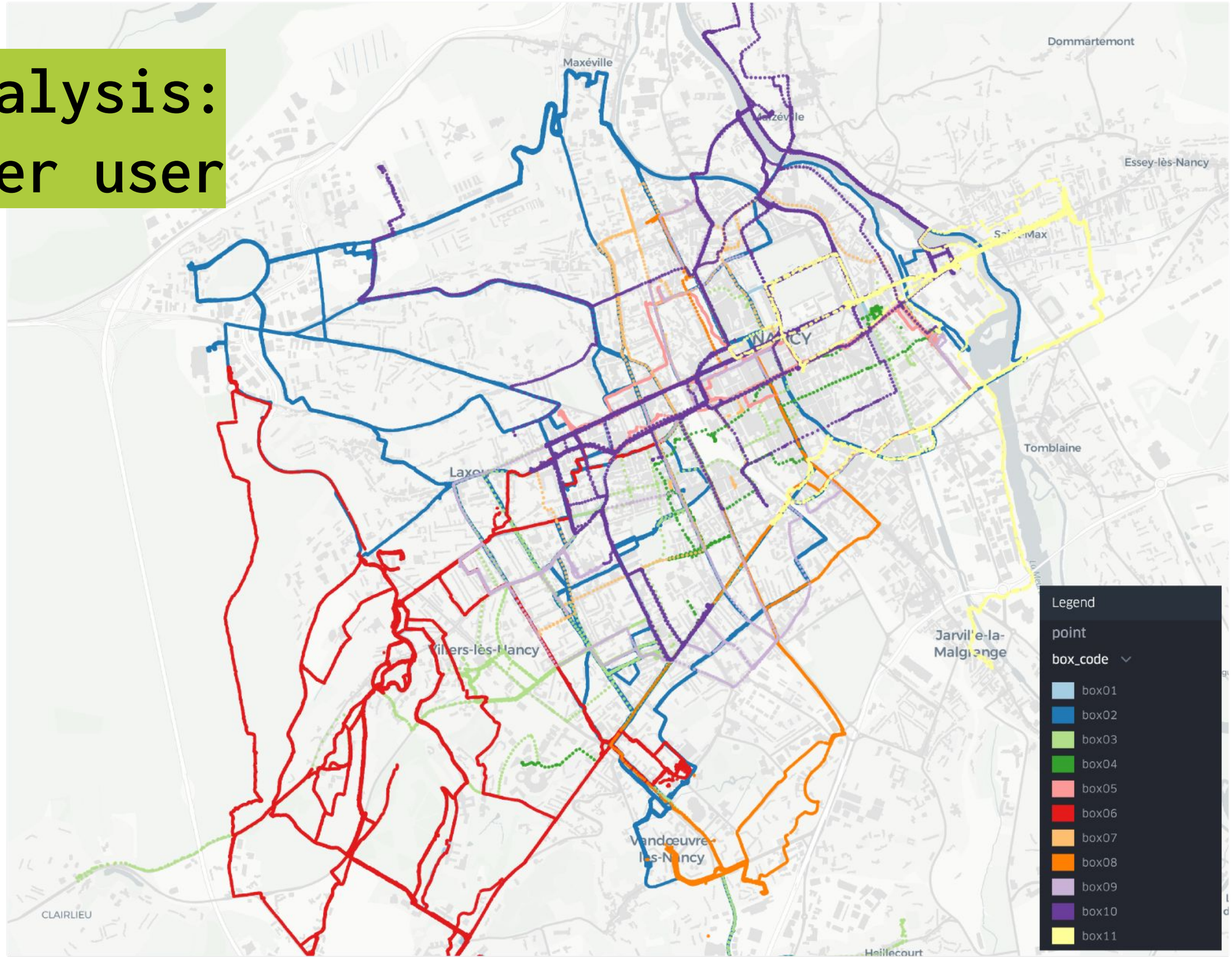
Campaign numbers

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Period: February 11 - Mai 20 / 2026

- 10 users / sensor boxes participated in the campaign
- 357 cycling trips recorded
- 1.234,74 km cycled in total
  (estimated from GPS movement, excluding a few unrealistic GPS jumps)
- 193.328 data points collected
- 84 hours recorded

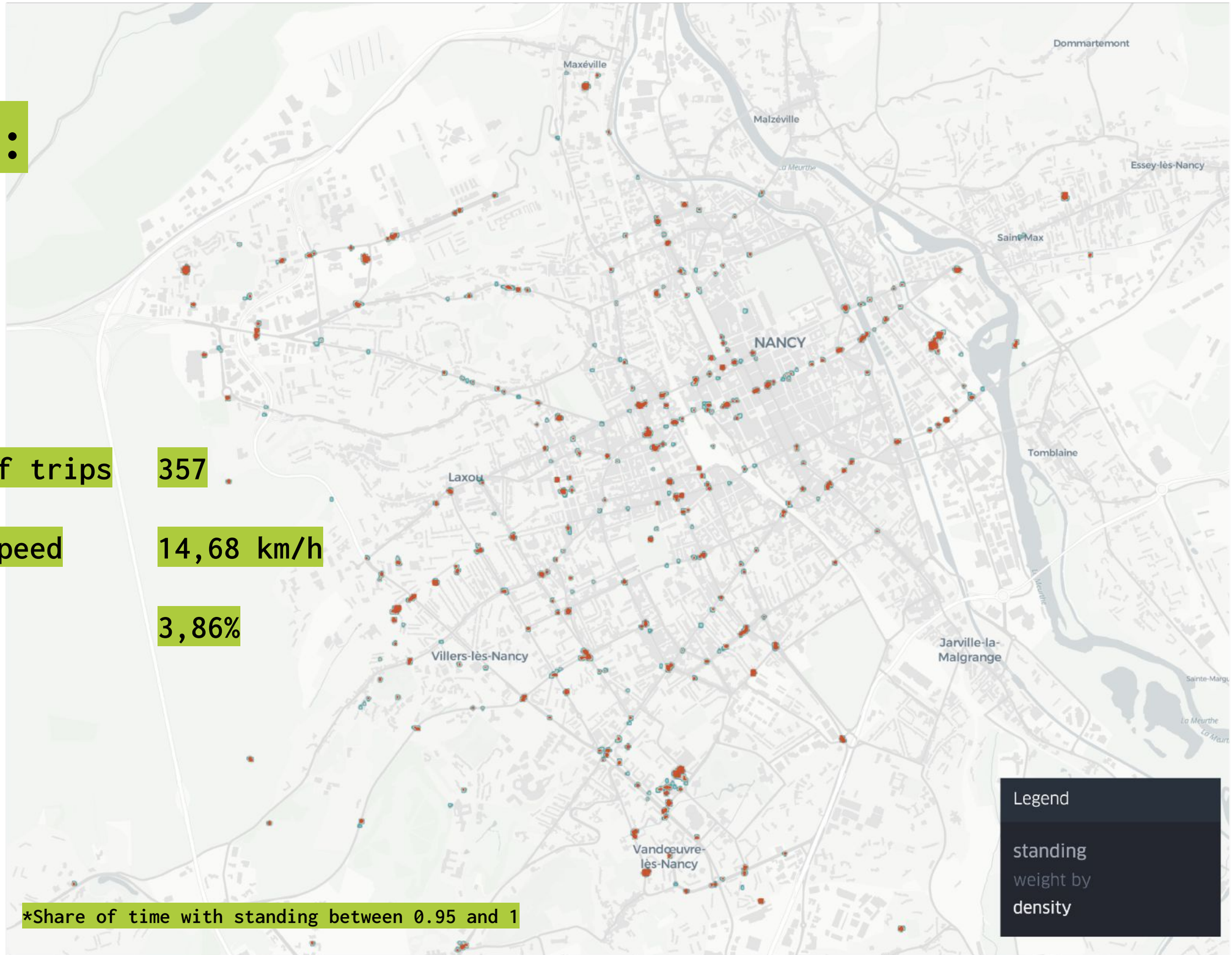
Data analysis: Rides per user



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Data analysis: Standing Hotspots

Total number of trips	357
Average trip speed	14,68 km/h
Standing*	3,86%



*Share of time with standing between 0.95 and 1

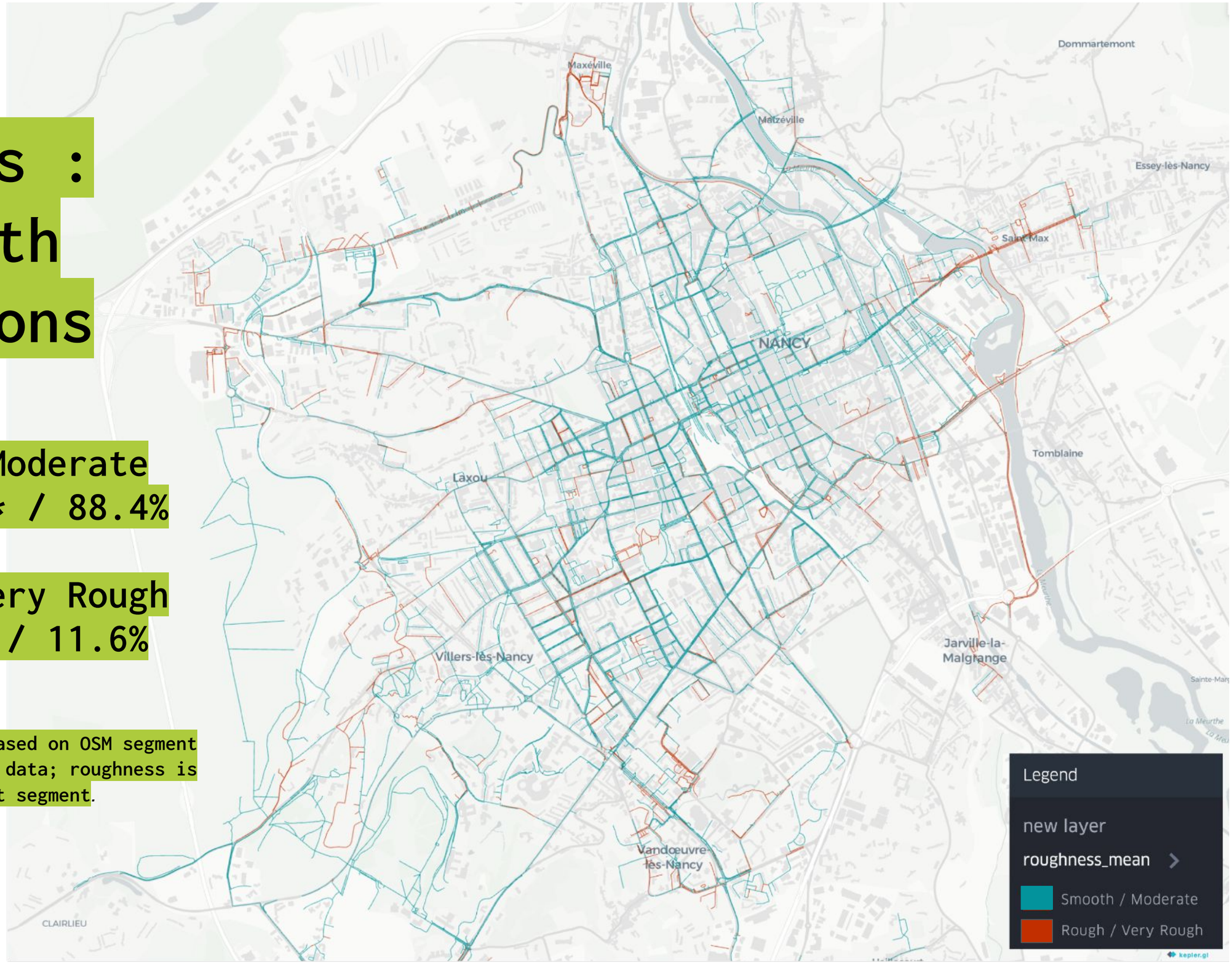
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Data analysis : bike path conditions

Smooth + Moderate
442.26 km* / 88.4%

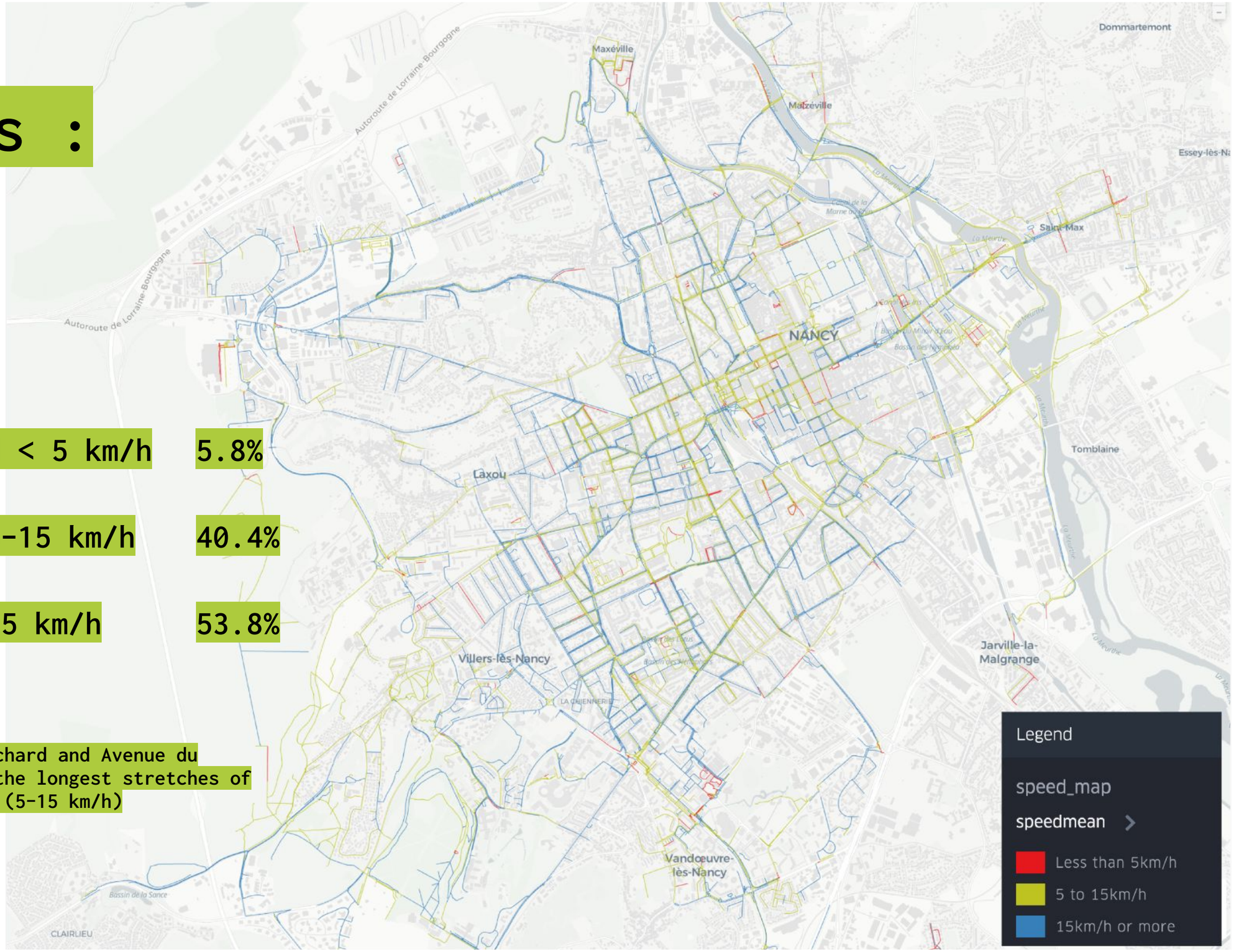
Rough + Very Rough
58.20 km* / 11.6%

*Percentages are based on OSM segment length with sensor data; roughness is averaged per street segment.



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Data analysis : Speed



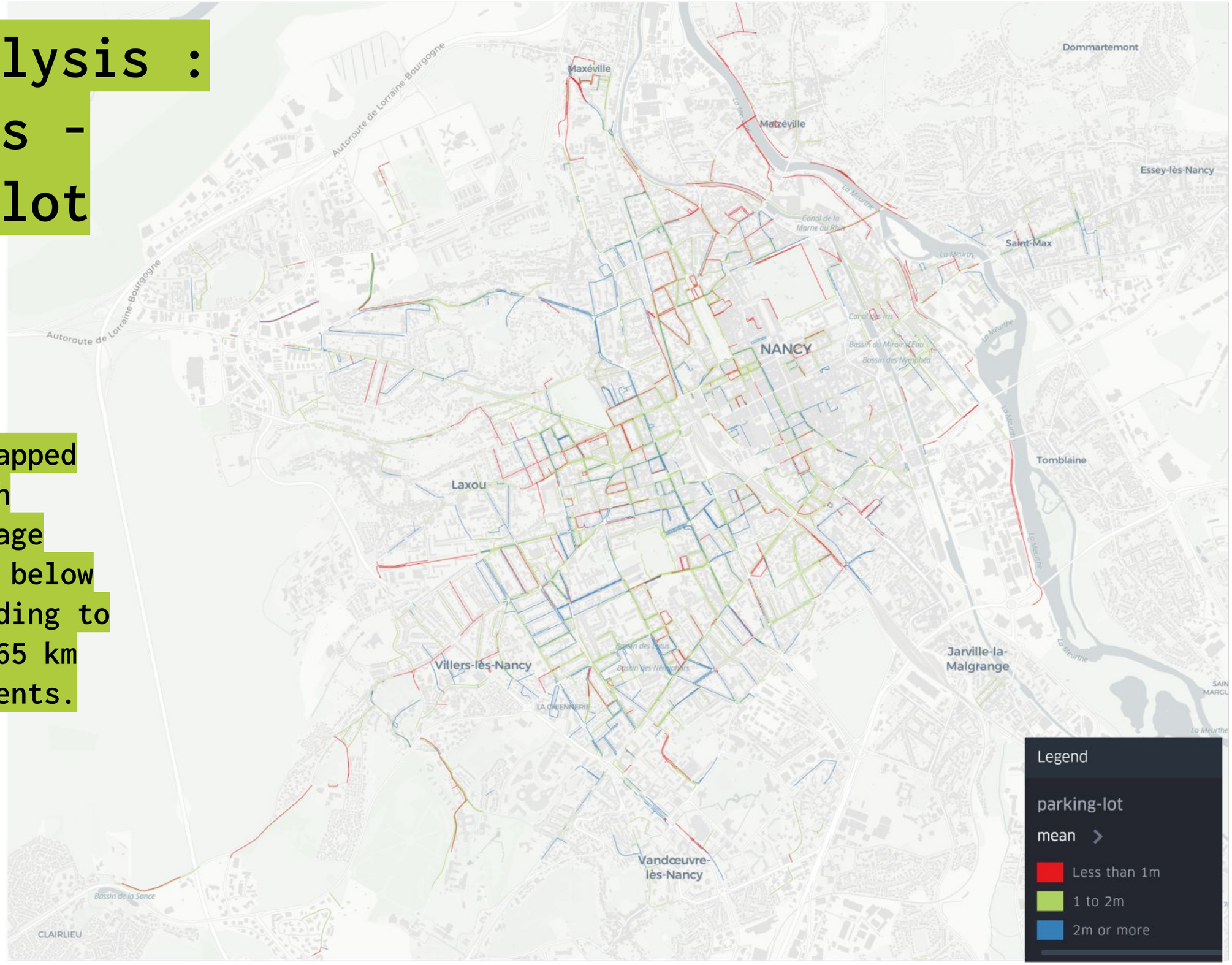
very low speed < 5 km/h	5.8%
Reduced flow 5-15 km/h	40.4%
Fluid flow > 15 km/h	53.8%

Ps:Avenue Raymond Pinchard and Avenue du Général Leclerc show the longest stretches of reduced cycling speed (5-15 km/h)

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Data analysis : Corridors - parking lot

Near on-street parking,
23.4% of the mapped corridor length showed an average corridor width below 1 m, corresponding to approximately 65 km of street segments.



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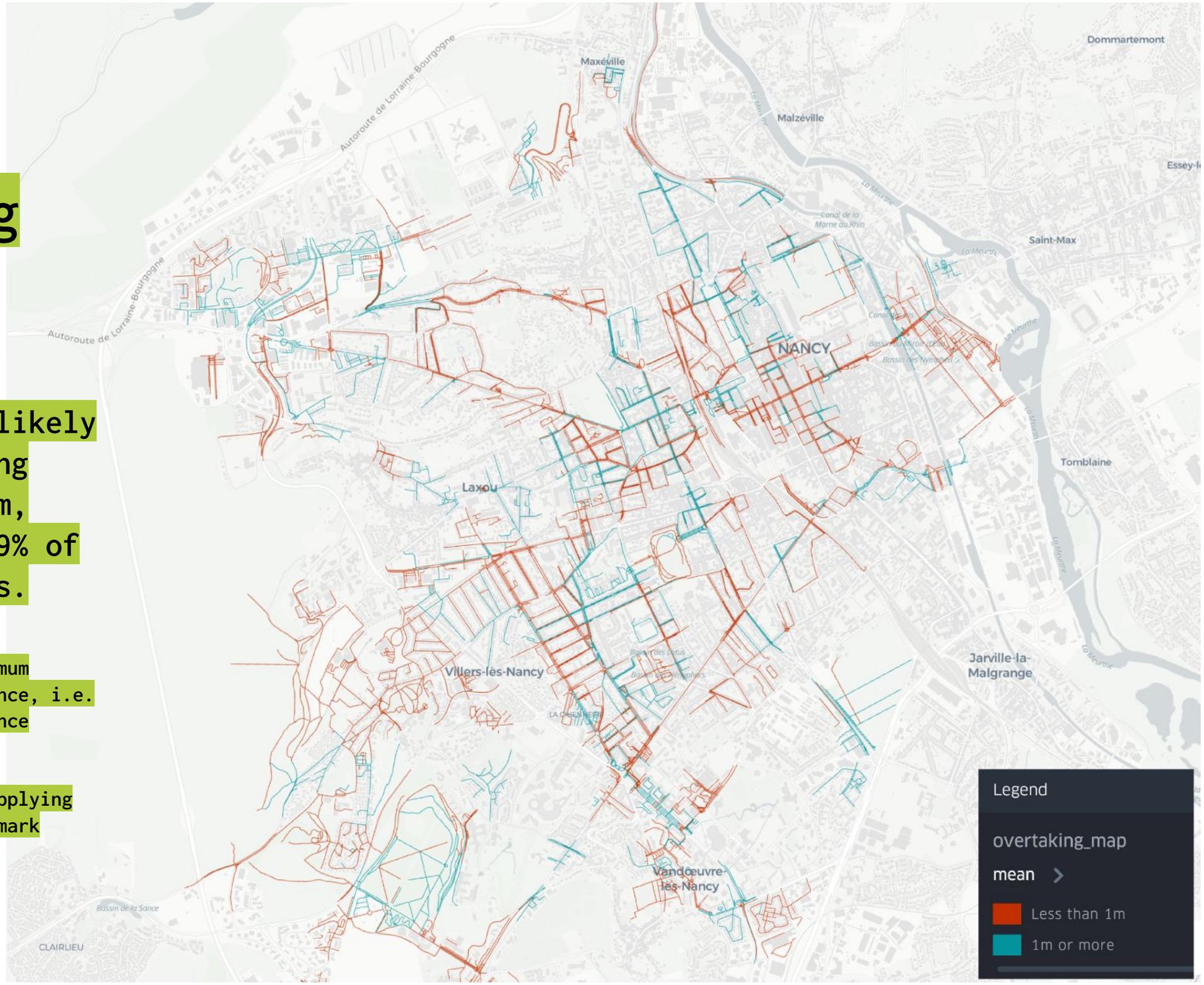
Data analysis: Overtaking distance

In general, 392 likely true close-passing events* below 1 m, representing 62.9% of overtaking events.

*Values refer to the minimum recorded overtaking distance, i.e. the closest passing distance measured for each event

*Estimated events after applying the Porto precision benchmark (47.1%).

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Data analysis : Overtaking distance

Rank	Street	Segments with at least one event < 1 m	Mean overtaking distance across those segments
1	Avenue de la Libération	29	72.5 cm
2	Avenue du Général Leclerc	23	77.4 cm
3	Boulevard de l'Europe	23	95.3 cm
4	Avenue de Bourgogne	22	78.2 cm
5	Rue Raymond Poincaré	18	75.7 cm

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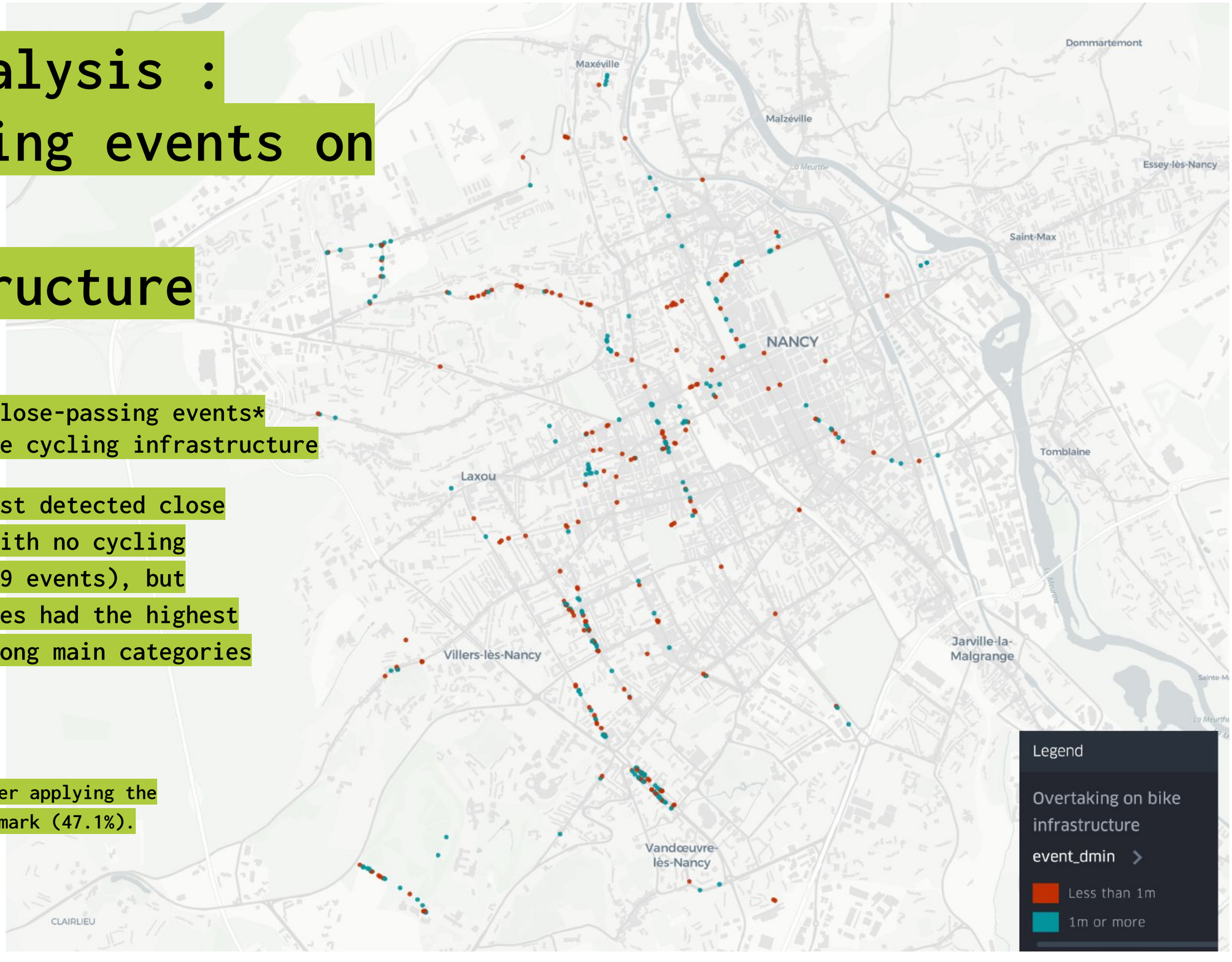
Data analysis : Overtaking events on cycling infrastructure

102 likely true close-passing events* below 1.0 m on the cycling infrastructure

Ps: Below 1 m, most detected close passes occurred with no cycling infrastructure (79 events), but painted cycle lanes had the highest critical share among main categories (65.2%).

*Estimated events after applying the Porto precision benchmark (47.1%).

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Data analysis: Overtaking events on cycling infrastructure

Right-side infrastructure type	*Events below 1 m	Total *events	Share below 1 m
AUCUN	38	79	48.2%
BANDE CYCLABLE	34	53	65.2%
PISTE CYCLABLE	15	37	40.5%
AMENAGEMENT MIXTE PIETON VELO HORS VOIE VERTE	7	16	42.4%
COULOIR BUS + VELO	6	15	37.5%
DOUBLE SENS CYCLABLE	2	3	66.7%
DOUBLE SENS CYCLABLE PISTE	1	1	33.3%

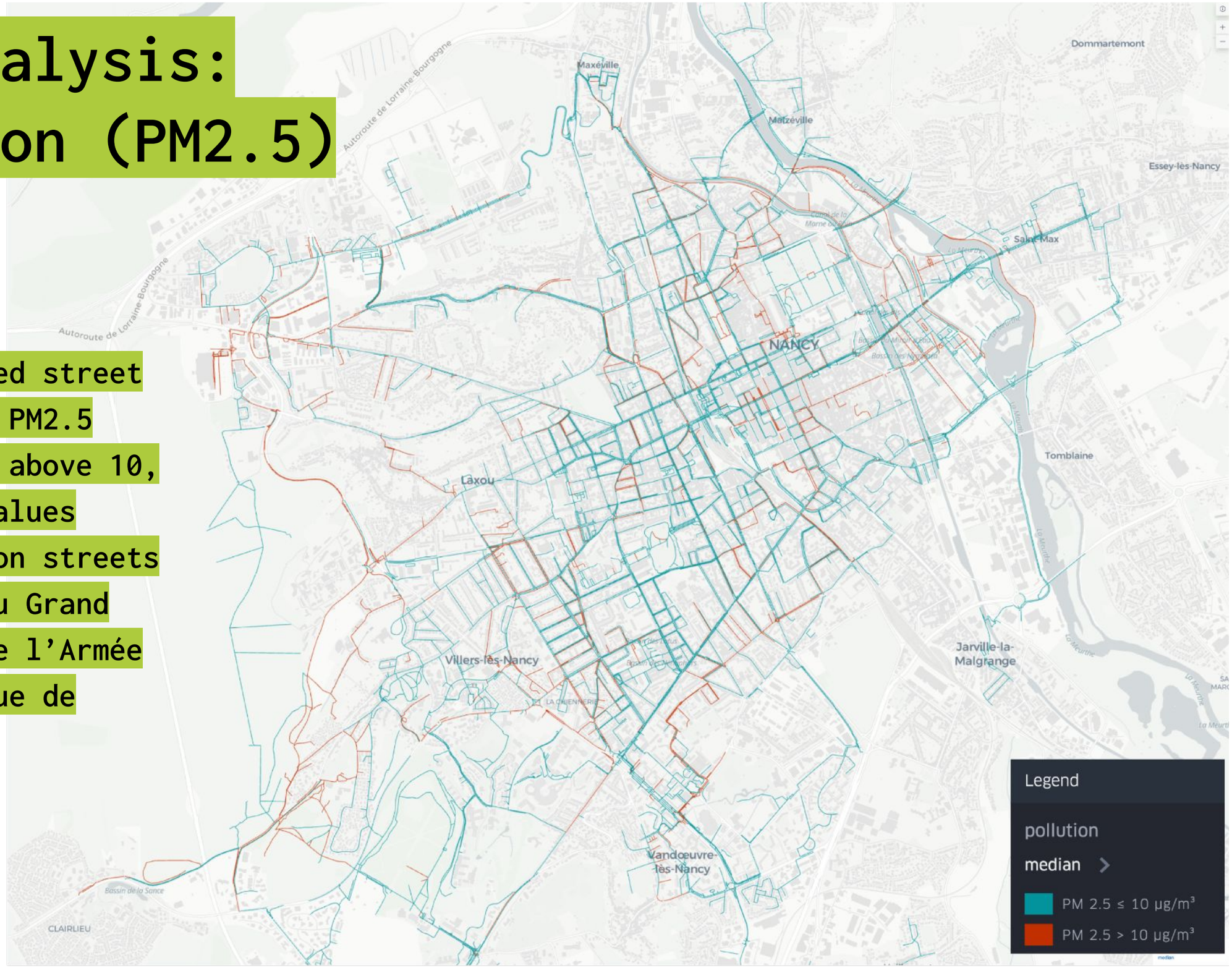
Below 1 m, most detected close passes occurred with no cycling infrastructure (79 events), but painted cycle lanes had the highest critical share among main categories (65.2%).

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*Estimated events after applying the Porto precision benchmark (47.1%).

Data analysis: Pollution (PM2.5)

19.2% of mapped street length showed PM2.5 median values above 10, with higher values concentrated on streets such as Rue du Grand Verger, Rue de l’Armée Patton, and Rue de Maréville.



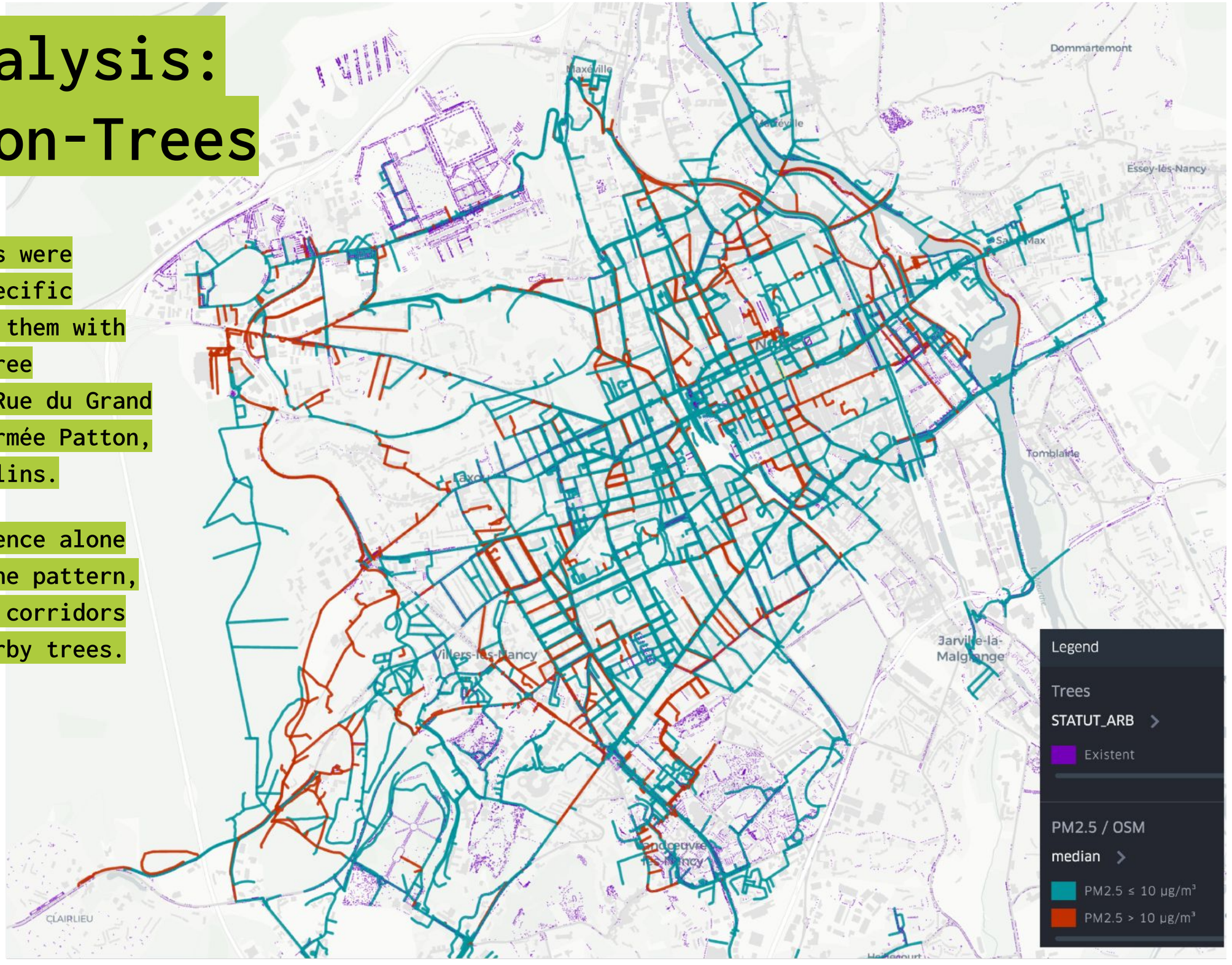
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Data analysis: Pollution-Trees

Higher PM2.5 values were concentrated on specific corridors, many of them with low or no nearby tree presence, such as Rue du Grand Verger, Rue de l’Armée Patton, and Rue des Tiercelins.

However, tree presence alone does not explain the pattern, as some high-PM2.5 corridors also have many nearby trees.

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Our perspective

We propose to collectively discuss and further develop Hypothesis #2

H2 • Cycling safety emerges from interactions between mobility modes

Cycling safety is shaped not only by infrastructure itself, but also by the dynamic interactions between cyclists, cars, pedestrians, and micro-mobility users.

Findings

- 102 probable overtaking events were detected with a distance below 1 m. Among all analysed overtaking events, painted cycle lanes showed the highest critical share, with 65.2% of events below this threshold.
- **Overall, 62.9% of detected overtaking events were below the critical threshold of 1 m.**
- **Near on-street parking, 23.4% of mapped corridor length had an average width below 1 m, covering approximately 65 km of street segments.**

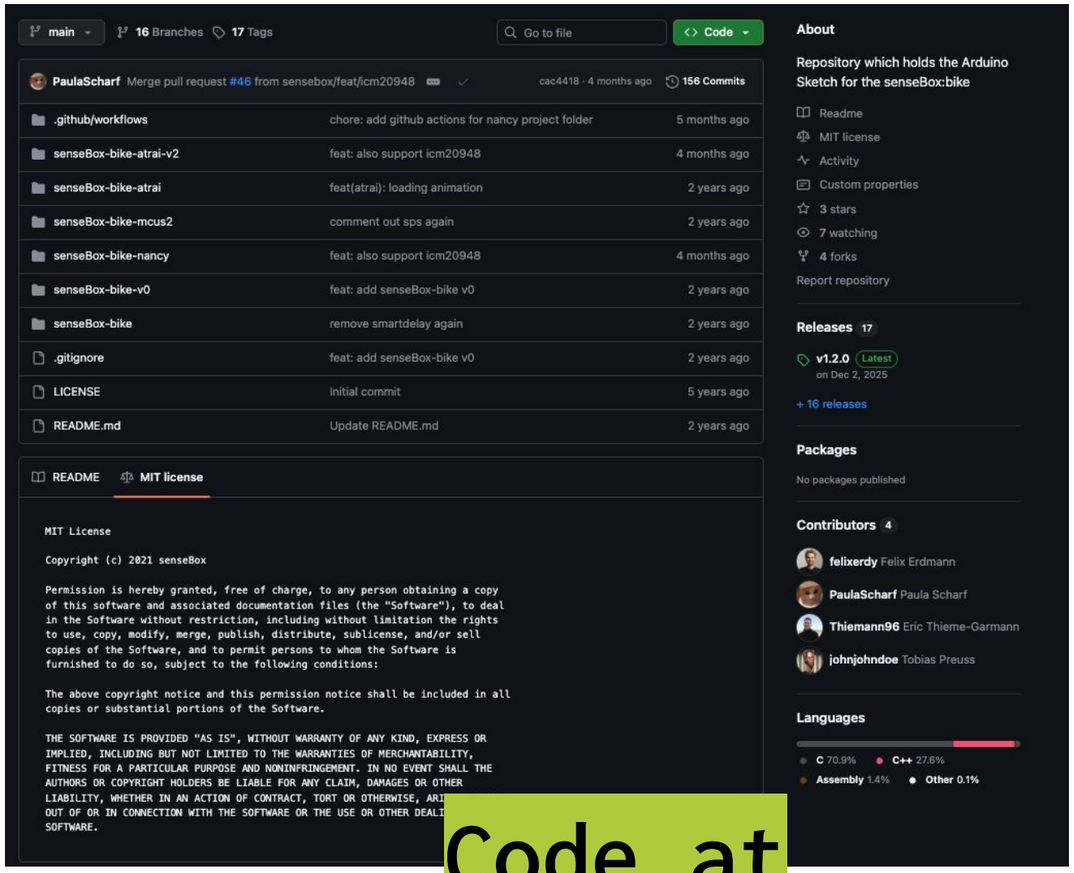
Implication

These findings support the legitimacy of advocating for 1.5 m+ overtaking distance and safer spatial separation between cars and cyclists as a relevant public safety measure.

Next step

This can serve as the starting point for the DATA-RIDE Workshop in Nancy (01.06.2026), where these findings and hypotheses can be collectively discussed, refined, and transformed into the basis for an advocacy campaign

Hardware and open source information



Code at github under MIT License



PDF with the assembly manual

Hardware and open source information

box_id	trip_id	timestamp_qgis	latitude	longitude	trip_direction	roughness_index	distance_left	overtaking_manoeuvre	distance_right	finedust_pm1	finedust_pm10	finedust_pm2_5	finedust_pm4	speed	standing
box_01	1	2026-02-11 18:53:48	48.6940066	6.1996575	outbound	5.25	34.0	0.89	53.0	0.5	1.01	0.74	0.93	0.0	0.85
box_01	1	2026-02-11 18:53:54	48.6939951	6.1996471	outbound	1.0	31.0	0.98	34.0	0.72	1.43	1.06	1.31	0.0	0.96
box_01	1	2026-02-11 18:54:08	48.6939215	6.199503	outbound	33.5	64.0	0.09	32.0	0.76	1.5	1.11	1.38	0.65	0.03
box_01	1	2026-02-11 18:54:11	48.6939459	6.1995565	outbound	1.25	73.0	0.0	107.0	1.19	2.35	1.74	2.16	0.18	0.95
box_01	1	2026-02-11 18:54:13	48.6939847	6.1995826	outbound	0.0	76.0	0.68	107.0					0.0	
box_01	1	2026-02-11 18:54:14	48.6939787	6.1995982	outbound	0.75	77.0	0.76	104.0	1.66	3.25	2.42	2.99	0.0	0.98
box_01	1	2026-02-11 18:54:15	48.6939231	6.1995338	outbound	0.0	68.0	0.0	114.0					0.0	
box_01	1	2026-02-11 18:54:18	48.6939551	6.1995659	outbound	3.5	72.0	0.0	97.0	1.76	3.45	2.56	3.16	0.0	0.9
box_01	1	2026-02-11 18:54:21	48.6939612	6.1995752	outbound	0.0	68.0	0.54	91.0					0.0	
box_01	1	2026-02-11 18:54:22	48.6939647	6.1995761	outbound	0.0	74.0	0.54	116.0					0.0	
box_01	1	2026-02-11 18:54:23	48.693966	6.1995749	outbound	5.0	86.0	0.01							
box_01	1	2026-02-11 18:54:24	48.6939734	6.1995538	outbound	0.0	115.0	0.03							
box_01	1	2026-02-11 18:54:26	48.6939855	6.1995163	outbound	5.75	47.0	0.01						1.02	0.82
box_01	1	2026-02-11 18:54:28	48.6939731	6.1995251	outbound	0.0	35.0	0.01						0.0	
box_01	1	2026-02-11 18:54:31	48.6939707	6.19952	outbound	0.0	122.0	0.01						0.0	
box_01	1	2026-02-11 18:54:34	48.6939709	6.1995154	outbound	0.0	35.0	0.02							
box_01	1	2026-02-11 18:54:37	48.6939884	6.1995246	outbound	0.0	151.0	0.02							
box_01	1	2026-02-11 18:54:41	48.6939943	6.199523	outbound	0.0	37.0	0.02						0.29	
box_01	1	2026-02-11 18:54:43	48.6939932	6.1995157	outbound	0.25	112.0	0.02					4.55	0.28	0.99

de-identified for public sharing by removing device identifiers, trip identifiers and timestamps.

Merci

Danke

Thanks

Obrigada

